IoT Project Task Assignment List

11 Members | 2 Raspberry Pis | 1 Week Duration

Task 1: Smartphone Sensor Data Collection

Duration: Days 1-3 | **3 Members**

🟦 MEMBER 1: Mobile App Development 🟦

- Develop React Native sensor app OR configure SensorLog
- Stream real-time sensor data to Pi
- Handle accelerometer, gyroscope, GPS data
- Test with mock server first, then integrate with Pi

Tools: Laptop, smartphone, Expo/React Native
Pi Required: Testing only (Day 3)

🟥 MEMBER 2: Pi Data Reception ⚠️ Pi #1 Required 🟥

- Python script on Raspberry Pi to receive data
- Database setup and data storage on Pi
- Real-time data validation and error handling
- Ensure continuous data flow management

Tools: Raspberry Pi #1, Python, SQLite
Pi Required: Days 1-3 (Full time)

🟨 MEMBER 3: Data Protocol & Validation 🟨

- Design data formats and communication protocols
- Create mock data generators for team testing
- Data cleaning and preprocessing pipelines
- Connection management systems

Tools: Laptop, Python
Pi Required: None

Task 2: Data Processing and Visualization

Duration: Days 2-5 | **4 Members**

🟩 MEMBER 4: Activity Recognition 🟩

- Process accelerometer data for activity detection
- Walking, running, sitting detection algorithms
- Movement pattern analysis and step counting
- Generate activity insights and metrics

Tools: Laptop, Python (NumPy, SciPy)
Pi Required: Testing only (Day 5)

🟪 MEMBER 5: GPS & Location Analytics 🟪

- GPS route mapping and tracking
- Speed and distance calculations
- Location-based insights and analytics
- Mapping integration (Folium/OpenStreetMap)

Tools: Laptop, Python (Folium, Geopy)
Pi Required: Testing only (Day 5)

🟫 MEMBER 6: 3D Orientation Tracking 🟫

- Process gyroscope data for phone orientation
- 3D rotation and tilt calculations
- Real-time orientation visualization
- Orientation change tracking algorithms

Tools: Laptop, Python (NumPy, Matplotlib 3D)
Pi Required: Testing only (Day 5)

🟧 MEMBER 7: Display & Visualization ⚠️ Pi #2 Required 🟧

- Create Matplotlib/Plotly visualizations
- Set up HDMI monitor display on Raspberry Pi
- Real-time graph updates and dashboard layout
- Monitor display optimization

Tools: Raspberry Pi #2, HDMI Monitor, Python
Pi Required: Days 3-5 (Display setup and testing)

Task 3: Web Interface for Remote Monitoring

Duration: Days 3-6 | **3 Members**

🔵 MEMBER 8: Flask Backend 🔵

- Develop Flask web application and REST API
- Real-time data serving and WebSocket support
- Database integration and data endpoints
- API design and implementation

Tools: Laptop, Python (Flask, SQLite)
Pi Required: Deployment only (Day 6)

🔶 MEMBER 9: Frontend Dashboard 🔶

- HTML/CSS/JavaScript web interface
- Real-time charts and interactive components
- Responsive design for multiple devices
- Dashboard UI/UX design

Tools: Laptop, HTML/CSS/JS, Chart.js
Pi Required: None

🔷 MEMBER 10: System Integration 🔷

- Deploy web interface to Pi network
- Connect all system components
- Cross-platform testing and integration
- Network accessibility configuration

Tools: Both Raspberry Pis for integration
Pi Required: Days 5-6 (Integration and deployment)

Task 4: Demonstration and Report

Duration: Days 5-7 | **1 Member**

🌟 MEMBER 11: Demo & Documentation ⚠️ Both Pis Required 🌟

- Complete system demonstration preparation
- Project documentation and report writing
- Demo video recording and presentation setup
- Troubleshooting guides and system documentation

Tools: Both Raspberry Pis, documentation tools
Pi Required: Days 6-7 (Full system demonstration)

Pi Usage Schedule

Pi #1 (Data Collection Focus)

- **Days 1-3:** Member 2 (Data reception setup)
- **Days 4-5:** Available for testing
- **Days 6-7:** Task 4 integration and demo

Pi #2 (Display & Processing Focus)

- **Days 1-2:** Setup and preparation
- **Days 3-5:** Member 7 (Monitor display setup)
- **Days 6-7:** Task 4 integration and demo

Development Requirements

Laptop Development (9 Members)

Members 1, 3, 4, 5, 6, 8, 9, 10

- Can work independently without Pi hardware
- Use mock data and local servers for testing
- Deploy/integrate with Pi only when needed

Pi Required Tasks (2 Members + Integration)

Member 2: Data reception on Pi #1
Member 7: Display setup on Pi #2
Member 11: Complete system demo (both Pis)

Weekly Timeline

Day 1: Foundation setup, data collection begins
Day 2: Core development, processing algorithms
Day 3: Parallel development, display setup
Day 4: Advanced development, web interface
Day 5: Integration preparation, testing
Day 6: System integration, deployment
Day 7: Final demo and documentation

Success Deliverables

✅ **Task 1:** Working smartphone → Pi data flow
✅ **Task 2:** Real-time visualizations on Pi monitor
✅ **Task 3:** Web interface accessible from network
✅ **Task 4:** Complete system demonstration + report

🎯 CHOOSE YOUR TASK - MEMBER SELECTION

💻 LAPTOP ONLY (Easy Access - No Pi Needed)

- 🟦 **MEMBER 1:** Mobile App Development
- 🟨 **MEMBER 3:** Data Protocol & Validation
- 🟩 **MEMBER 4:** Activity Recognition
- 🟪 **MEMBER 5:** GPS & Location Analytics
- 🟫 **MEMBER 6:** 3D Orientation Tracking
- 🔵 **MEMBER 8:** Flask Backend
- 🔶 **MEMBER 9:** Frontend Dashboard

⚠️ RASPBERRY PI REQUIRED (Limited Access)

- 🟥 **MEMBER 2:** Pi Data Reception (Pi #1 - Days 1-3)
- 🟧 **MEMBER 7:** Display & Visualization (Pi #2 - Days 3-5)
- 🔷 **MEMBER 10:** System Integration (Both Pis - Days 5-6)
- 🌟 **MEMBER 11:** Demo & Documentation (Both Pis - Days 6-7)

Instructor: Bhavik Gandhi
Project Duration: 1 Week
Team Size: 11 Members